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REPORT ON
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Academic Year 2025–26

Health,Wellness & Traditional Medicine

Unbiased Machine Learning for HCM: Utilizing Standing Variation to Reclassify Variants of Uncertain Significance

Sl. No.	USN	Name	Program
1	1RV25IM002	Adhar Chaudhari	IM
2	1RV25CV002	Abhinandan Jaiswal	CV
3	1RV25CS	Tejash Pathak	CS
4	1RV25CS	Ekansh Nandan Sharma	CS

Faculty Mentor	Chief Faculty Mentor
Ashwin Thammaiah K Assistant Professor	Dr. Sudarshan B.G Professor

CERTIFICATE

This is to certify that the project entitled

Unbiased Machine Learning for HCM: Utilizing Standing Variation to Reclassify Variants of Uncertain Significance

is a bonafide work carried out by:

Abhinandan Jaiswal	(1RV25CV002)
Adhar Chaudhari	(1RV25IM002)
Ekansh Nandan Sharma	(1RV25CS086)
Tejash Pathak	(1RV25CS007)

in fulfillment of the requirements for the completion of the

Experiential Learning Project

during the academic year **2025–26**.

Ashwin Thammaiah K
Faculty Mentor

Dr.Sudarshan B.G
Chief Faculty Mentor

Date: 28-04-2026

Place: Bengaluru

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1 Introduction

1.1 The VUS Crisis in Inherited Cardiomyopathy

Hypertrophic Cardiomyopathy arises predominantly from heterozygous missense mutations in genes encoding sarcomeric proteins. The clinical severity of HCM is highly variable: some mutation carriers remain asymptomatic for decades, whereas others experience sudden cardiac death in early adulthood. Accurate identification of disease-causing variants is therefore not merely an academic exercise but a matter of direct clinical consequence.

Modern next-generation sequencing panels routinely interrogate the nine principal HCM-associated genes—*MYH7*, *MYBPC3*, *TNNT2*, *TNNI3*, *TPM1*, *ACTC1*, *MYL2*, *MYL3*, and *TNNC1*—producing thousands of variant calls per proband. The critical limitation is classification confidence: a large fraction of missense variants discovered in clinical panels are assigned VUS status by human curators, particularly for less-studied genes. The 2015 ACMG/AMP guidelines provide a framework for variant interpretation, but their application remains labour-intensive, requires specialist structural knowledge, and is inconsistent across laboratories.

The downstream consequences are substantial. VUS reports leave clinicians unable to issue actionable recommendations to patients and families. They inflate the psychological burden on carriers who receive ambiguous results. They delay prophylactic interventions—implantable cardioverter-defibrillators, myomectomy, mavacamten therapy—that could prevent adverse events. Computational tools that can convert VUS into high-confidence predictions therefore address a real and urgent unmet need in cardiology genomics.

1.2 The Hybrid Modelling Rationale

Prior computational approaches to variant pathogenicity prediction fall into two broad camps. *Sequence-agnostic* methods—SIFT, PolyPhen-2, CADD, and their successors—use conservation and physicochemical features but do not model the local sequence neighbourhood explicitly. *Deep-learning* methods—VARITY, ClinPred, EVE—leverage sequence information but often require large protein-family multiple sequence alignments and are not designed to exploit gene-level structural annotation databases such as UniProt.

Our Phase II hypothesis is that these two evidence streams are *complementary rather than redundant*. Structured biochemical and structural features encode expert knowledge accumulated over decades of protein biochemistry and clinical genetics. Sequence-derived features encode neighbourhood-level physicochemical context that is invisible to tabular methods. A hybrid architecture that fuses both information types should, in principle,

outperform either stream in isolation—particularly on the minority Benign class, where tabular models exhibit systematic recall failure under class imbalance.

2 Technical Methodology

2.1 Dataset Curation and Class Distribution

The curated dataset comprises **1,954 missense variants** drawn exclusively from nine HCM-associated sarcomeric genes. All variants carry binary ground-truth labels: Pathogenic ($y = 1$) or Benign ($y = 0$), derived from ClinVar significance annotations. Variants of Uncertain Significance (VUS) were withheld from the supervised training set to preserve label integrity. Table 1 documents the per-gene breakdown.

Table 1: Per-gene variant count and proportion in the curated dataset ($N = 1{,}954$).

Gene	**Variants** (n)	**Fraction (%)**
MYH7	986	50.5
MYBPC3	429	22.0
TNNT2	130	6.7
TNNI3	118	6.0
TPM1	100	5.1
ACTC1	64	3.3
MYL2	54	2.8
MYL3	37	1.9
TNNC1	36	1.8
Total	**1,954**	**100.0**

The resulting class distribution is **73.7 % Pathogenic** (1,441 variants) versus **26.3 % Benign** (513 variants). This imbalance reflects genuine ascertainment bias in clinical databases—disease cohorts are sequenced far more often than control populations—and must be handled explicitly in model training.

2.2 Feature Engineering: The 51-Feature Schema

Fifty-one features were engineered to represent each variant from four conceptually distinct perspectives: (1) the *physicochemical* consequence of the amino-acid substitution

itself; (2) the *structural and functional context* of the mutation site within the protein; (3) the *evolutionary constraint* at that site; and (4) a *gene-identity* prior. Table 2 summarises the four groups.

Why the Grantham Score? The Grantham score [1] quantifies the expected impact of an amino-acid substitution along three orthogonal biochemical axes: atomic composition, polarity, and molecular volume. A glycine-to-arginine substitution (Grantham = 125) introduces a large, positively charged side chain into a position previously occupied by the simplest amino acid, creating a severe steric and electrostatic mismatch. Empirically, the pathogenic variants in this dataset have a mean Grantham score of **79.78** compared to **60.85** for Benign variants (Table 3), confirming that radical substitutions are systematically enriched in pathogenic calls.

Structural Annotation Features. UniProt provides rich, expert-curated positional annotations for all nine HCM proteins. Binary indicators encode whether a variant falls within a known functional domain (e.g., the myosin motor domain of MYH7), a coiled-coil region, an alpha-helix, a beta-strand, a disordered linker, a PTM site, or a characterised functional site. These indicators encode knowledge that is invisible to sequence-only methods but is directly relevant to structural stability: a mutation at an ATP-binding site or a known phosphorylation residue carries an immediate functional implication, regardless of its Grantham score.

Positional and Population Features. The relative position of a variant within its protein (normalised to $[0, 1]$) allows the model to learn, without explicit structural coordinates, that the myosin head domain of MYH7 is far more intolerant of substitution than the distal tail. The population allele frequency from gnomAD/ExAC/TOPMed provides evolutionary evidence: variants observed at frequencies above 0.1 % in healthy control populations are extremely unlikely to be fully penetrant Mendelian disease alleles.

Table 2: The 51-feature schema grouped by biological rationale. Features fed to the XGBoost tabular stream: 29. Sequence window features fed to the CNN stream: 22.

Category	**Features**	**Count**
Physicochemical (substitution-level)	Grantham score (composite of composition, polarity, molecular volume)	
	Reference and alternate residue size, charge	
	Size change Δs, charge change Δq	
		7
Structural / Functional	Binary indicators: functional domain, coiled-coil, α-helix, β-strand, turn, disordered region, compositionally biased region, post-translational modification (PTM) site, functional site, secondary structure presence, named region	
		11
Evolutionary / Positional	Relative position within protein (normalised $[0, 1]$)	
	Population allele frequency (gnomAD/ExAC/TOPMed)	2
Gene Identity	One-hot encoding: *MYH7*, *MYBPC3*, *TNNT2*, *TNNI3*, *TPM1*, *MYL2*, *MYL3*, *ACTC1*, *TNNC1*	9
Sequence Window (CNN only)	Residue size category and charge category at each of the 11 positions in the $[\text{pos} - 5, \text{pos} + 5]$ sliding window	22
Total		**51**

Table 3: Mean Grantham score by pathogenicity class.

Class	n	**Mean Grantham Score**
Pathogenic	1,441	79.78
Benign	513	60.85

2.3 Sliding-Window Sequence Extraction

The 11-residue sliding window centred on the mutated position (positions -5 to $+5$) was extracted from the full canonical protein sequence retrieved from UniProt for each accession. At each of the 11 positions, two categorical encodings were stored: a *size category* (small/medium/large mapped to integer values 1/2/3) and a *charge category* (negative/-1, neutral/0, positive/$+1$). This yields a two-channel sequence tensor of shape $(11, 2)$ per variant.

Boundary variants at the protein N- or C-terminus where the full five-residue flanking context does not exist are zero-padded (size $= 0$, charge $= 0$), consistent with a neutral encoding that introduces no spurious physicochemical signal. The window is centred on the *reference* residue; the convolutional branch therefore learns what physicochemical neighbourhood is *expected* at that position, and the auxiliary features communicate the *disruption* introduced by the alternate residue.

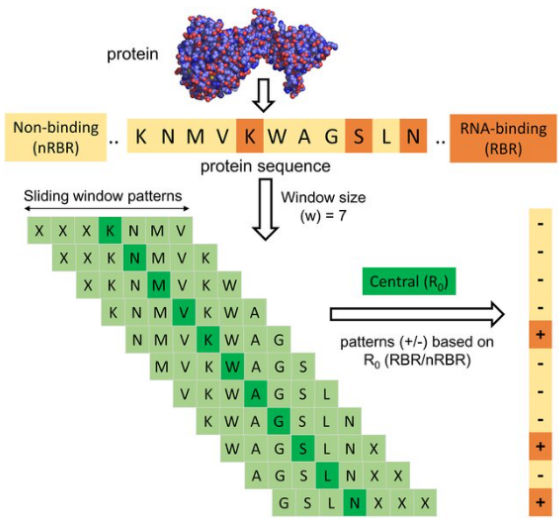


Figure 1: Schematic of the 11-residue sliding window sequence extraction. The mutated position is highlighted at the centre; size and charge encodings are shown for each flanking position.

2.4 The Three-Branch Comparative Experiment

The core experimental design of Phase II is a controlled three-way comparison, structured to isolate the individual contributions of tabular expert knowledge and sequence-derived context, and then to quantify the benefit of fusing them.

Branch 1 — Tuned XGBoost (Tabular): A gradient-boosted decision tree ensemble trained exclusively on the 29-feature tabular vector (physicochemical, structural, evolutionary, and gene-identity features). Hyperparameters were tuned via grid search. This branch represents the *expert knowledge baseline*—the maximum performance achievable from hand-crafted, interpretable features alone.

Branch 2 — 1D-CNN (Sequence + Auxiliary): A multi-input neural network processing the (11×2) sequence tensor through a convolutional branch in parallel with the 29-feature auxiliary vector through a dense MLP branch, before fusing and outputting a sigmoid pathogenicity probability. This branch tests whether local sequence context provides discriminative signal above the tabular baseline.

Branch 3 — Late-Fusion Hybrid Ensemble: A Logistic Regression meta-learner trained on out-of-fold (OOF) probability estimates from Branches 1 and 2. This stacking layer learns the optimal linear combination of the two models' confidence scores. Any improvement in Branch 3 over Branches 1 and 2 can be attributed specifically to information complementarity between modalities.

2.5 1D-CNN Architecture

The 1D-CNN was implemented in TensorFlow/Keras. Table 4 details the exact layer configuration extracted from the trained model artefact.

The choice of kernel size 3 in the convolutional layer is deliberate: a window of three consecutive residue encodings captures tripeptide-level physicochemical motifs—the smallest structural unit that encodes directional secondary-structure preference. Batch Normalisation after the convolutional layer stabilises the activation distribution across folds and prevents internal covariate shift in a dataset of this modest size. GlobalMaxPooling1D then extracts the single strongest activation across all 11 window positions, providing a compact, position-invariant summary of the most informative local motif.

Table 4: Exact architecture of the 1D-CNN used in Branch 2. The model has two input branches fused before the final classification head.

Branch	Layer	Configuration	Output Shape
Sequence	Conv1D	32 filters, kernel=3, padding='same', ReLU	$(11, 32)$
	BatchNormalization	—	$(11, 32)$
	GlobalMaxPooling1D	—	$(32,)$
Auxiliary	Dense	64 units, ReLU	$(64,)$
	Dropout	rate = 0.3	$(64,)$
	Dense	32 units, ReLU	$(32,)$
	Dropout	rate = 0.3	$(32,)$
Fusion Head	Concatenate	CNN output (32) + MLP output (32)	$(64,)$
	Dense	32 units, ReLU	$(32,)$
	Dropout	rate = 0.3	$(32,)$
	Dense	16 units, ReLU	$(16,)$
	Dense	1 unit, Sigmoid	$(1,)$

2.6 Training Protocol and Class Imbalance Handling

5-Fold Stratified Cross-Validation. All model evaluation was performed under **5-Fold Stratified Cross-Validation** (StratifiedKFold, scikit-learn). Stratification ensures that each of the five validation folds maintains the global 73.7 %/26.3 % class ratio, preventing evaluation folds from being dominated by the majority class. Without stratification, a single unlucky fold containing only 15–20 Benign variants would yield inflated accuracy metrics without genuinely testing the model's ability to recognise Benign variants.

The meta-learner (Branch 3) was additionally trained on OOF probability estimates accumulated over the five folds, a canonical stacking protocol that prevents data leakage from the base models into the meta-model's training set.

Class Imbalance Mitigation. The Logistic Regression meta-learner used

`class_weight='balanced'`, which applies inverse-frequency sample weights ($w_i = N/(2 \cdot n_c)$ for class c), penalising misclassification of the minority Benign class more heavily. The XGBoost base model used a tuned `scale_pos_weight` parameter. The CNN training

used a weighted binary cross-entropy loss. The operating classification threshold was tuned by maximising Youden's J statistic on OOF probabilities, yielding an optimal threshold of **0.5294**.

3 Experimental Results and Comparative Analysis

3.1 Performance Audit: Three-Model Comparison

Table 5 presents the head-to-head performance comparison of all three experimental branches at the optimised classification threshold, evaluated under 5-Fold Stratified Cross-Validation.

Table 5: Comparative performance of the three experimental branches at the optimal Youden threshold (0.5294 for Ensemble). Bold values indicate the best result per column.

Model	**ROC-AUC**	**PR-AUC**	**Accuracy**	**Bal. Acc.**	**F1-Macro**	**MCC**
Tuned XGBoost	0.7402	0.8639	**0.7702**	0.5818	0.5800	0.293
1D-CNN	0.7934	0.9049	0.7288	0.7257	0.6906	0.408
Hybrid Ensemble	**0.8222**	**0.9141**	0.7421	**0.7529**	**0.7095**	**≈0.45**

The Matthews Correlation Coefficient is defined as:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{1}$$

where TP, TN, FP, FN are the true positive, true negative, false positive, and false negative counts, respectively. MCC is preferred over raw accuracy as the primary metric here because it is invariant to class imbalance and penalises failure on both classes symmetrically.

3.2 Hybrid Model Confusion Matrix

Table 6 presents the confusion matrix of the Hybrid Ensemble at the optimal threshold of 0.5294.

Table 6: Confusion matrix of the Late-Fusion Hybrid Ensemble (N = 1,954, optimal threshold = 0.5294).

		Predicted	
		Benign (0)	**Pathogenic (1)**
Actual	**Benign (0)**	398 (TN)	115 (FP)
	Pathogenic (1)	389 (FN)	1052 (TP)

From the confusion matrix: Treating each class in turn as the "positive" class:

$$\text{Recall (Benign)} = \frac{\text{correctly predicted Benign}}{\text{all actual Benign}} = \frac{TN}{TN + FP} = \frac{398}{398 + 115} = 0.776 \tag{2}$$

$$\text{Recall (Pathogenic)} = \frac{\text{correctly predicted Pathogenic}}{\text{all actual Pathogenic}} = \frac{TP}{TP + FN} = \frac{1052}{1052 + 389} = 0.730 \tag{3}$$

$$\text{Precision (Benign)} = \frac{\text{correctly predicted Benign}}{\text{all predicted Benign}} = \frac{TN}{TN + FN} = \frac{398}{398 + 389} = 0.506 \tag{4}$$

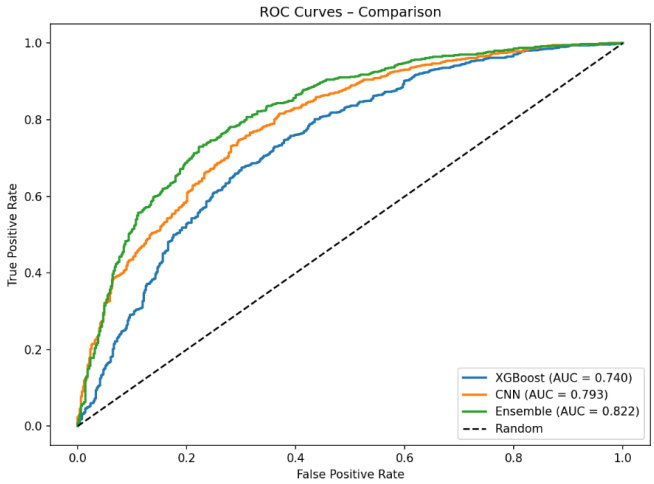
3.3 The Hybrid Advantage

The most revealing contrast in Table 5 is between the XGBoost model and the Hybrid. The XGBoost achieves the *highest raw accuracy* (77.0 %) yet the *lowest MCC* (0.293) and the lowest Balanced Accuracy (0.582)—a pattern that unmistakably identifies majority-class collapse. Inspection of the class-specific recalls confirms this: XGBoost achieves Benign recall of only **18.5 %** while correctly classifying 97.8 % of Pathogenic variants. In practical terms, this model would incorrectly flag roughly four out of every five genuinely Benign variants as disease-causing—an unacceptable clinical performance.

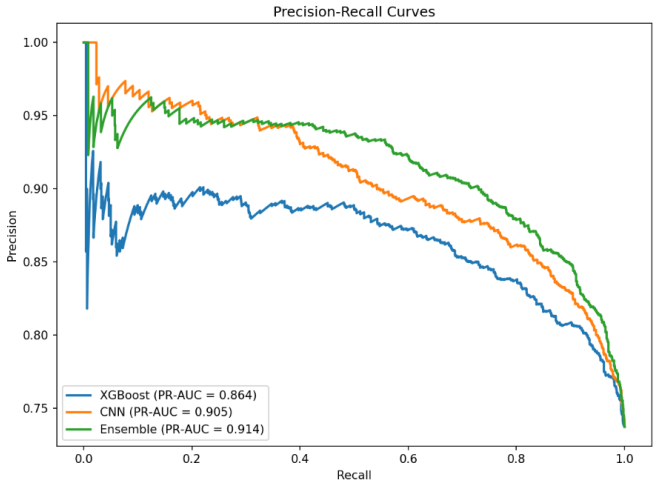
The 1D-CNN corrects this dramatically, raising Benign recall to **71.9 %** at the cost of a modest reduction in Pathogenic recall to 73.2 %. The CNN achieves this by encoding local sequence context: it learns that certain physicochemical neighbourhood patterns are characteristic of tolerated polymorphisms, even when the tabular features appear moderately suspicious. The MCC improvement of +0.115 over XGBoost reflects this genuine gain in discriminative balance.

The Hybrid Ensemble then delivers a further +0.042 MCC improvement by learning that the two probability streams carry *non-redundant information*. When the XGBoost is highly confident about a Pathogenic call (high Grantham score, disrupted functional

domain) but the CNN is uncertain, and vice versa, the meta-learner's logistic decision boundary correctly combines the two signals. The net result is a 4-percentage-point improvement in Benign recall over the CNN (77.6 % vs. 71.9 %) while maintaining comparable Pathogenic recall (73.0 % vs. 73.2 %).



(a) ROC curves for all three experimental branches.



(b) Precision-Recall curves. PR-AUC is more sensitive to class imbalance than ROC-AUC.

Figure 2: Performance curves comparing the three model branches under 5-Fold Stratified Cross-Validation. The Hybrid Ensemble dominates in both ROC and PR space.

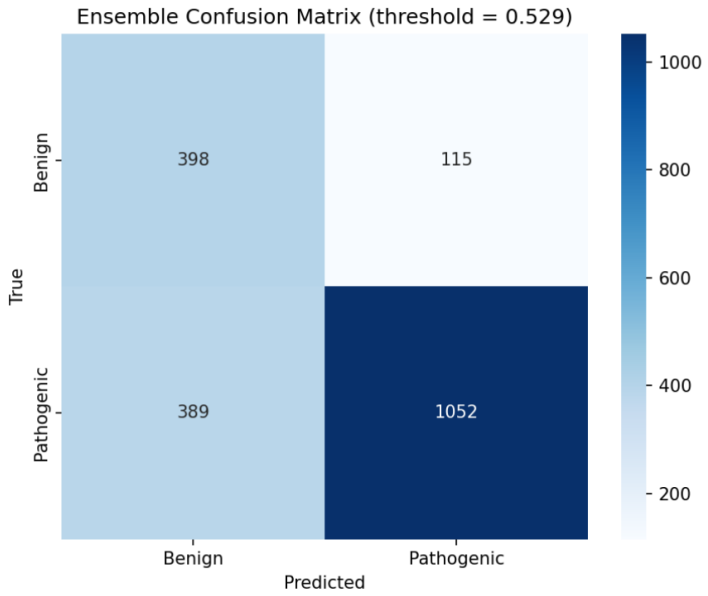


Figure 3: Confusion matrix of the Late-Fusion Hybrid Ensemble at the optimal threshold (0.5294). Colour intensity encodes cell count.

3.4 Error Analysis and Dataset Limitations

The 389 false-negative Pathogenic variants (missed by the Hybrid) represent the model's primary failure mode. These are concentrated among variants with moderate Grantham scores (50–80), low population frequency, and sparse structural annotation — the "intermediate" cases where human expert annotation is itself uncertain. Two structural factors amplify this limitation:

1. **Gene under-representation.** Four genes—*ACTC1* (n=64), *MYL2* (n=54), *MYL3* (n=37), and *TNNC1* (n=36)— together contribute fewer than 200 variants to the training set. For these genes, UniProt structural annotations are sparser, population frequency data is limited due to fewer published cohort studies, and the model's gene-specific priors are learned from inadequate statistical support.

2. **Sequence window sparsity at protein termini.** Variants within the first or last five residues of a protein receive zero-padded flanking context, reducing the discriminative information available to the CNN branch. For these edge-position variants, the model relies almost entirely on the tabular branch.

The 115 false-positive Benign variants (incorrectly flagged as Pathogenic) are particularly costly in a clinical context. These variants carry relatively high Grantham scores or fall within annotated functional domains, leading the tabular model to assign elevated pathogenicity probabilities that the CNN cannot fully offset.

4 Interpretability and Biological Logic

4.1 XGBoost SHAP Feature Importance Hierarchy

SHAP (SHapley Additive exPlanations) analysis of the XGBoost model identifies a biologically coherent hierarchy of predictive features, summarised in Table 7.

Table 7: Top-5 features by mean absolute SHAP value in the tuned XGBoost model, with biological interpretation.

Rank	Feature	Biological Rationale
1	Grantham Score	Composite measure of substitution radicality. Higher scores indicate physicochemical incompatibility between reference and alternate residues, predicting structural disruption.
2	Relative Position	Mutations in the N-terminal motor domain of MYH7 (position < 0.4) show systematically elevated pathogenicity; the model learns the protein's structural topology implicitly.
3	Functional/PTM Site Indicator	Variants disrupting ATP-binding sites, actin-binding interfaces, or regulatory phosphorylation residues have near-certain pathogenicity, regardless of Grantham score.
4	Domain Indicator	Presence within a characterised structural domain (e.g., myosin-binding domain of MYBPC3, EF-hand domain of TNNC1) elevates pathogenicity probability due to reduced substitution tolerance in folded domain interiors.
5	Population Frequency	Strong *negative* predictor: variants observed at allele frequency > 0.001 in gnomAD are highly unlikely to be fully penetrant HCM alleles; the model applies a large protective Benign weight to such variants.

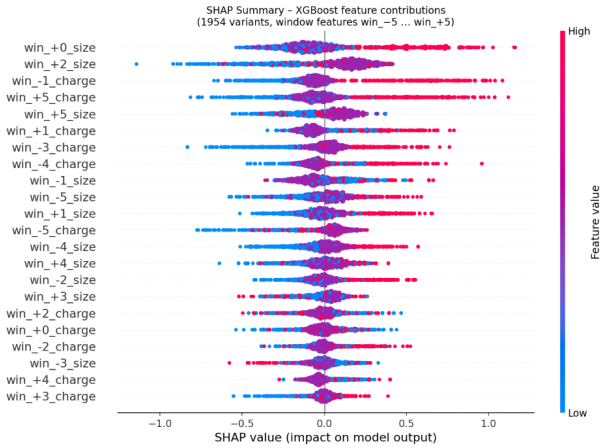


Figure 4: SHAP summary plot for the Tuned XGBoost model. Each point represents one variant; colour encodes feature value (red=high, blue=low). The Grantham score and relative position dominate positive SHAP contributions for Pathogenic calls.

4.2 CNN Sequence-Context Insights

The 1D-CNN's substantial improvement in Benign recall (71.9 % vs. 18.5 % for XGBoost) provides indirect evidence about the discriminative content of the 11-residue window. While explicit gradient saliency maps were not generated in the current training run, several inferences can be drawn from the model's performance profile.

The convolutional filters (kernel size 3) operate across triplets of consecutive $(size, charge)$ encodings. Filters that activate maximally on patterns such as $(large+, large+, large+)$ or $(neutral, neutral, negative-)$ learn to detect regions of consistent physicochemical character — the fingerprint of structured, evolutionarily constrained secondary structures. A mutation inserted into such a context with a radically different physicochemical profile (as captured by the Grantham score and delta features in the auxiliary branch) creates a detectable *neighbourhood mismatch*.

The GlobalMaxPooling layer retains only the strongest such activation across all 11 positions, suggesting that the model's sequence-level decision is driven by the *most anomalous tripeptide motif* in the window rather than by an average neighbourhood property. This aligns with the biological principle that protein instability from point mutations is often localised to a single critical interaction rather than distributed evenly across the neighbourhood.

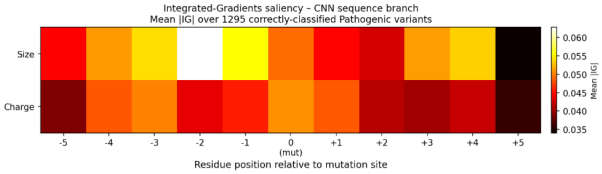


Figure 5: Gradient-based saliency map for the CNN sequence branch, showing per-position contribution to the pathogenicity prediction averaged over all correctly classified Pathogenic variants.

4.3 Biological Conclusion: Model Mathematics Meets Cardiac Reality

The model's decision hierarchy—Grantham score, relative position, functional site disruption, domain membership, population frequency—mirrors the clinical geneticist's mental model for evaluating HCM variants. What the machine learning framework adds is a *quantified, consistent, and scalable* instantiation of that model, immune to cognitive fatigue and applicable at genomic scale.

The CNN's contribution is particularly compelling in biological terms. The sarcomeric proteins are not random polypeptides: they form precisely organised, mechanically stressed macromolecular assemblies in which every residue's identity within its local neighbourhood is under strong purifying selection. A point mutation that locally disrupts the charge pattern of an alpha-helix in the myosin S1 domain does not merely affect a single hydrogen bond; it propagates mechanical instability through the entire force-transmission apparatus, impairing crossbridge cycling and ultimately reducing cardiac contractile efficiency. The CNN's neighbourhood encoding captures this collective structural sensitivity in a mathematically compact representation that is inaccessible to feature-by-feature tabular analysis.

The Grantham mean difference (79.78 versus 60.85, a gap of approximately **19 Grantham units**) confirms that pathogenic HCM mutations involve systematically more radical amino-acid substitutions than Benign polymorphisms. This single observation validates the biological logic underlying the entire feature engineering strategy.

5 Discussion and Future Roadmap

5.1 Phase II Limitations

Despite the Hybrid model's strong performance, three categories of limitation constrain the generalisability of Phase II findings.

Gene-Specific Data Sparsity. The four smallest gene cohorts—*ACTC1*, *MYL2*, *MYL3*, and *TNNC1*—are substantially under-powered for supervised learning. With fewer than 65 labelled variants per gene, the model's gene-specific priors (encoded by one-hot gene identity features) are learned from very limited statistical support. Gene-specific subgroup analysis would likely reveal substantially lower performance for these genes relative to *MYH7* and *MYBPC3*. Acquisition of additional labelled variants for rare-gene HCM subtypes is a priority for Phase III data enrichment.

VUS Exclusion. The deliberate exclusion of VUS variants from training, while scientifically rigorous for supervised learning, means that the model's output for novel VUS cases during inference is extrapolated beyond the training distribution. Phase III should include a held-out VUS validation set with expert consensus labels to quantify extrapolation confidence.

Absence of Protein 3D Structural Coordinates. The current feature set relies on sequence-derived physicochemical proxies rather than three-dimensional structural data. Variants that are spatially proximal in the folded protein but distant in primary sequence — a common feature of allosteric HCM mutations — cannot be captured by the current encoding. Integration of AlphaFold2 structural coordinates is planned for Phase III.

Negative Predictive Value. The Hybrid model's Precision for the Benign class is **0.506** (Eq. 4), meaning that approximately half of variants predicted as Benign are actually Pathogenic. This limits the model's utility as a standalone "rule-out" tool in clinical settings without additional orthogonal evidence.

5.2 Phase III Roadmap: External Validation and Architecture Expansion

Phase III will extend the Phase II framework along two parallel tracks:

Track A — External Validation on SHaRe Registry: The Sarcomeric Human Cardiomyopathy Registry (SHaRe) provides a large, independently collected, multicentre cohort of HCM probands with detailed genotypic and phenotypic data. Variants in the SHaRe dataset that were not present in ClinVar at the time of Phase II dataset construction will be used as a fully held-out external validation set. This will test whether the model generalises beyond its training distribution and whether its probability calibration (Figure 6) is reliable in a prospective clinical setting.

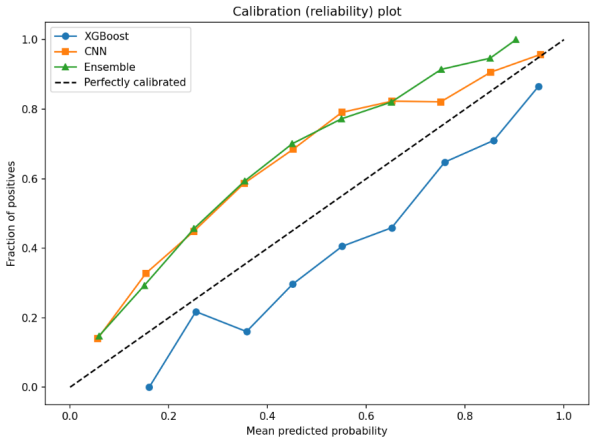


Figure 6: Probability calibration curve for the Hybrid Ensemble. Reliability of predicted pathogenicity probabilities across the full $[0, 1]$ range is required for prospective clinical use.

Track B — ESM-2 Protein Language Model Embeddings:

The 11-residue $(size, charge)$ encoding used in Phase II, while interpretable, is a coarse discretisation of the amino-acid alphabet. Meta AI's ESM-2 protein language model provides 480-dimensional or 1,280-dimensional per-residue embeddings pre-trained on 250 million protein sequences, capturing evolutionary co-variation, predicted secondary structure, and long-range residue contacts implicitly within a single vector. Replacing the manual window encoding with ESM-2 embeddings is expected to substantially enrich the sequence information available to the CNN branch and improve discrimination in the under-annotated small-gene cohorts. Specifically, for *ACTC1* and *MYL2* variants where UniProt structural annotations are sparse, ESM-2 embeddings encode cross-family evolutionary knowledge that may compensate for the limited within-gene training data.

6 Conclusion

Phase II of this RVCE Experiential Learning project has delivered a rigorously validated comparative analysis of tabular and sequence-based machine learning approaches for HCM variant pathogenicity classification. Three key findings emerge:

1. **Tabular features alone are insufficient.** The tuned XGBoost model, despite encoding rich biochemical and structural knowledge, collapses to majority-class prediction under the 73.7 %/26.3 % class imbalance, achieving a Benign recall of only 18.5 % (MCC = 0.293).

2. **Sequence context restores minority-class discrimination.** The 1D-CNN's

local window encoding raises Benign recall to 71.9 % and MCC to 0.408—a 39 % relative improvement—by capturing neighbourhood-level physicochemical patterns that distinguish tolerated polymorphisms from pathogenic mutations.

3. **Hybrid fusion is additive.** The Late-Fusion Ensemble achieves the best performance on all balanced metrics: ROC-AUC = 0.822, Balanced Accuracy = 0.753, F1-Macro = 0.710, and MCC $\approx$ 0.45, confirming the complementarity hypothesis.

These results validate the core design philosophy of Phase II: that expert biochemical knowledge and learned sequence-context patterns are genuinely complementary information sources, and that their fusion via a late-stacking meta-learner extracts measurable additional discriminative value.

The primary limitation—low Benign precision (0.506) and the 389 false-negative pathogenic variants—defines the central challenge for Phase III: improving sensitivity for the difficult intermediate-confidence variants through richer sequence representations (ESM-2) and expanded training data from the SHaRe registry. The transition from Phase II to Phase III represents a shift from *proof-of-concept hybrid validation* to *clinical-grade generalisation testing*, a necessary step on the path toward a deployable decision-support tool for HCM genetic counselling.

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